

CONCEPT SUMMARY: Solving Rational Equations

Words

A rational equation is an equation that contains a rational expression. To solve, identify the domain for the variable. Then multiply both sides of the equation by a common denominator and solve. An extraneous solution is a solution that is not valid because that value is excluded from the domain of the original equation.

Algebra

$$\frac{1}{x} + \frac{2}{x} = \frac{1}{6} \quad \text{Domain: } x \neq 0$$

$$6x \left(\frac{1}{x} + \frac{2}{x} \right) = 6x \left(\frac{1}{6} \right)$$

$$6 + 12 = x$$

$$18 = x$$

The domain includes $x = 18$, so the solution to

$$\frac{x^2 + 4}{x - 1} = \frac{5}{x - 1} \quad \text{Domain: } x \neq 1$$

$$(x - 1) \left(\frac{x^2 + 4}{x - 1} \right) = (x - 1) \left(\frac{5}{x - 1} \right)$$

$$x^2 + 4 = 5$$

$$x^2 = 1$$

$$x = \pm 1$$

The domain does not include $x = 1$, so 1 is an

It does include $x = -1$, so the solution to the